Executive Summary

Risk Assessment for the Digitalization and Global Expansion of Pamper Pets

Purpose:

Digitalization and global expansion can yield benefits such as increased revenue, brand consolidation, and more. Still, this also exposes Pamper Pets to new risks that must be addressed to succeed in this endeavor. This document summarizes the probabilities of a “perfect order”, the risks of the supply chain, and recommendations to mitigate them, and prescribes a disaster-recovery architecture that guarantees 1-minute RPO and RTO.

Scope and method

First, a reliability block diagram (RBD) and Monte Carlo with a triangular distribution are used to calculate the probability endangering both the quality and availability of the product. These methods are used because:

- RDBs are used in different applications for calculating the probability of failure or unavailability and allow for representing the logic of the system states (Signoret and Leroy, 2021)
- Triangular distribution is used for generating different cases based on the external data collected. This distribution follows the Norden-Rayleigh distribution and is bounded (Jones, 2019). This allows us to have a good dataset for the Monte Carlo simulation.

To ensure proper decision-making and risk ranking across categories, MDCM techniques were used. Grey-TOPSIS, based on the work of Rathore et al. (2017), is chosen and shows the following benefits:

- Grey system incorporates the situation of fuzziness.
- Allows aggregation of opinion for a balanced and technical consensus
- Enable stakeholders to take appropriate risk responses.
- Assist decision-making to design a resilient supply chain.
- Validated with subject matter experts.

Key assumptions

The following are some assumptions for the data and results.

- The RDB used is a good simplification of the factors that affect a global supply chain.
- The data to calculate the probability of failure is based on multiple external reports because of the limited data on Pamper Pets
- The scale for risk assessment is obtained from Rathore et al.'s research.
- The scoring for risks is based on the Pamper Pets team's business knowledge.
- The different risks are obtained based on various reports, but the risk classification vary to consider the different Pamper Pet’s situation.
- The results show a snapshot in time; for this reason, it is necessary to perform a risk analysis at least once a year.

Quantitative results

Probability of supply chain and quality failure

A serial system is used to model the possibility that failures in automated warehouses, quality inspection, international shipping, and cyber risk could disrupt operations, affecting product availability and quality. Figure 1 shows that automated warehouses have an availability of 99.5% to 99.99% per year (Tong, 2025), with up to a 95% efficiency gain (PITD, 2022). Second, quality control depends on the type of sensors and control systems used; we estimate the failure rate of sensors in a redundant system at 97% to 99% per year. Third, international on-time-in-full (OTIF) rates vary by sector, ranging from 95% to 99% (DHL, 2020). Finally, there is a cyber risk that could disrupt Pamper Pets' supply chain; we discuss this in more detail in the next sections. The failure rate for the proposed architecture is 99.999856%.

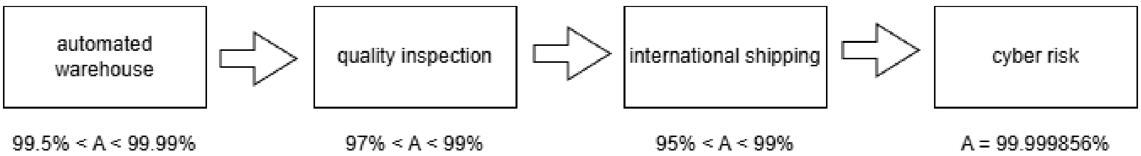

Figure 1: Reliability block diagram of the supply chain

Using the serial reliability formula based on the maximum values:

$$R_{sys} = \prod_{i=1}^{n} R_i = R_{warehouse} \times R_{quality} \times R_{shipping} \times R_{cyber}$$

$$R_{sys} = 0.9999 \times 0.99 \times 0.99 \times 0.99999856 \approx 0.98$$

The deterministic probability of a “Perfect Order” is 98%

RDB provides a snapshot but fails to account for the uncertainty and variance in international logistics. A Monte Carlo simulation is performed using a million iterations, using a triangular distribution for each variable. The code in Appendix 1 was used to run the simulation.

The results show that the 5th percentile is 93.34%, which is a substantial decrease compared to the RDB result. The 95th percentile is 96.3%, with a 2.96% difference from the 5th percentile, indicating that an automated, optimized supply chain can deliver low operational volatility.

Considering the HPIs, this still represents a high failure rate, as it means one out of 15 packages could arrive late or be damaged. There is a need to take a “white glove” approach to high-profile customers, as failure can be catastrophic for the company's reputation.

Risk Assessment

Scale for assessment

Table 1 shows the grey scale for the evaluation:

Categories risk		Probability of occurrence and impact		Increase in activity and cost	
Linguistic term	Grey number	Linguistic term	Grey number	Linguistic term	Grey number
Strategic	[0, 0.2]	Low	[0, 0.2]	No influence	[0, 0]
Mid-strategic-tactical	[0.2, 0.4]	Medium	[0.2, 0.4]	Low influence	[0, 0.2]
Tactical	[0.4, 0.6]	High	[0.4, 0.6]	Medium influence	[0.2, 0.5]
Mid-tactical-operational	[0.6, 0.8]	Very high	[0.6, 0.8]	High influence	[0.5, 0.8]
Operational	[0.8, 1]	Extreme	[0.8, 1]	Very high influence	[0.8, 1]

Table 1:Scale for assessment based on Rathore et al. (2017)

Risks

The following risks and categories were identified based on past research reports.

Category	Risk	Code
Supply Risk	Farmers' inability to supply	R1
	Non-availability of the procurement center	R2
	Poor quality of supply food	R3
	Communication Failure	R4
	Unreliable supplier	R5
	Natural disaster	R6
Storage and Transport Risk	Insufficient capacity of the warehouse	R7
	Poor handling while loading or unloading	R8

	Poor packaging or preservation	R9
	Unreliable or poorly managed inventory	R10
	In transit loss	R11
	Timely availability of vehicles	R12
Social risk	Malfunction in PDS	R13
	Theft or risk from outside	R14
	Lack of transparency on processes	R15
	Technological risk	R16
	Labor strike	R17
Demand risk	Sudden spike in demand	R18
	Demand Forecasting error	R19

Table 2: Risks based on Rathore et al. (2017), Behzadi et al. (2018), and Vlajic et al. (2012)

Assessment

Table 3 shows the scoring of the different risks based on the linguistic terms determined above.

	Category of risk	Probability of risk	Impact of risk	Increase in activity duration due to the given risk	Increase in cost due to given risk
R1	O	M	E	VH	VH
R2	TO	L	VH	M	M
R3	O	H	E	H	VH
R4	S	M	M	H	M
R5	T	M	M	M	M
R6	O	L	VH	VH	VH
R7	TO	L	M	M	H
R8	O	M	M	M	M
R9	TO	M	M	M	M
R10	T	H	M	H	M
R11	O	L	VH	VH	H
R12	TO	M	M	M	L
R13	TO	M	0.5	M	L
R14	0.5	L	VH	VH	M
R15	ST	H	VH	M	M
R16	ST	H	M	M	L
R17	TO	L	E	VH	VH
R18	S	M	M	H	L
R19	ST	M	M	H	M

Table 3: Assessment based on Pamper Pets information

Table 4 shows the calculation of numerical values and their denominator that will be used to normalize the values

	Category of risk	Probability of risk	Impact of risk	Increase in activity duration due to given risk	Increase in cost due to given risk
R1	0.967	0.267	0.967	0.967	0.967
R2	0.733	0.033	0.733	0.323	0.323
R3	0.967	0.5	0.967	0.677	0.967

R4	0.033	0.267	0.5	0.677	0.323
R5	0.5	0.267	0.5	0.323	0.323
R6	0.967	0.033	0.733	0.967	0.967
R7	0.733	0.033	0.5	0.323	0.677
R8	0.967	0.267	0.5	0.323	0.323
R9	0.733	0.267	0.5	0.323	0.323
R10	0.5	0.5	0.5	0.677	0.323
R11	0.967	0.033	0.733	0.967	0.677
R12	0.733	0.267	0.5	0.323	0.033
R13	0.733	0.267	0.5	0.323	0.033
R14	0.5	0.033	0.733	0.967	0.323
R15	0.267	0.5	0.733	0.033	0.323
R16	0.267	0.5	0.5	0.323	0.033
R17	0.733	0.033	0.967	0.967	0.967
R18	0.033	0.267	0.5	0.677	0.033
R19	0.267	0.267	0.5	0.677	0.323
denominator	2.9774526	1.2837971	2.87083821	2.793351213	2.3665018

Table 4: Grey values converted to numerical values

Table 5 shows the normalized values.

	Category of risk	Probability of risk	Impact of risk	Increase in activity duration due to given risk	Increase in cost due to given risk
R1	0.3247743	0.08967397	0.32477427	0.324774272	0.3247743
R2	0.2461836	0.0110833	0.2461836	0.108481996	0.108482
R3	0.3247743	0.16792879	0.32477427	0.227375576	0.3247743
R4	0.0110833	0.08967397	0.16792879	0.227375576	0.108482
R5	0.1679288	0.08967397	0.16792879	0.108481996	0.108482
R6	0.3247743	0.0110833	0.2461836	0.324774272	0.3247743
R7	0.2461836	0.0110833	0.16792879	0.108481996	0.2273756
R8	0.3247743	0.08967397	0.16792879	0.108481996	0.108482
R9	0.2461836	0.08967397	0.16792879	0.108481996	0.108482
R10	0.1679288	0.16792879	0.16792879	0.227375576	0.108482
R11	0.3247743	0.0110833	0.2461836	0.324774272	0.2273756
R12	0.2461836	0.08967397	0.16792879	0.108481996	0.0110833
R13	0.2461836	0.08967397	0.16792879	0.108481996	0.0110833
R14	0.1679288	0.0110833	0.2461836	0.324774272	0.108482
R15	0.089674	0.16792879	0.2461836	0.0110833	0.108482
R16	0.089674	0.16792879	0.16792879	0.108481996	0.0110833
R17	0.2461836	0.0110833	0.32477427	0.324774272	0.3247743
R18	0.0110833	0.08967397	0.16792879	0.227375576	0.0110833
R19	0.089674	0.08967397	0.16792879	0.227375576	0.108482

Table 5: Normalized values

Table 6 illustrates the final calculations to get the score denoted in P.

	Category	Probability	Impact	Increase in activity duration	Increase in cost due to the given risk	S+	S-	P
R01	0.3248	0.0897	0.3248	0.3248	0.3248	0.0783	0.5709	0.8795
R02	0.2462	0.0111	0.2462	0.1085	0.1085	0.3613	0.2835	0.4397
R03	0.3248	0.1679	0.3248	0.2274	0.3248	0.0974	0.5411	0.8475
R04	0.0111	0.0897	0.1679	0.2274	0.1085	0.4306	0.2499	0.3672
R05	0.1679	0.0897	0.1679	0.1085	0.1085	0.3859	0.2230	0.3663
R06	0.3248	0.0111	0.2462	0.3248	0.3248	0.1754	0.5489	0.7578
R07	0.2462	0.0111	0.1679	0.1085	0.2274	0.3341	0.3340	0.4999
R08	0.3248	0.0897	0.1679	0.1085	0.1085	0.3525	0.3515	0.4993
R09	0.2462	0.0897	0.1679	0.1085	0.1085	0.3612	0.2836	0.4398
R10	0.1679	0.1679	0.1679	0.2274	0.1085	0.3248	0.3248	0.5000
R11	0.3248	0.0111	0.2462	0.3248	0.2274	0.2007	0.4997	0.7135
R12	0.2462	0.0897	0.1679	0.1085	0.0111	0.4267	0.2663	0.3843
R13	0.2462	0.0897	0.1679	0.1085	0.0111	0.4267	0.2663	0.3843
R14	0.1679	0.0111	0.2462	0.3248	0.1085	0.3196	0.3723	0.5381
R15	0.0897	0.1679	0.2462	0.0111	0.1085	0.4546	0.2154	0.3215
R16	0.0897	0.1679	0.1679	0.1085	0.0111	0.4744	0.2007	0.2972
R17	0.2462	0.0111	0.3248	0.3248	0.3248	0.1754	0.5260	0.7499
R18	0.0111	0.0897	0.1679	0.2274	0.0111	0.4868	0.2301	0.3210
R19	0.0897	0.0897	0.1679	0.2274	0.1085	0.3772	0.2620	0.4099
V-	0.3248	0.1679	0.3248	0.3248	0.3248			
V+	0.0111	0.0111	0.1679	0.0111	0.0111			

Table 6: Values for calculating ranking

Table 7 shows that different risks are ranked by their TOPSIS score. The main risks are related to supply chain availability and social risks.

	P
R01	0.879460381
R03	0.847456307
R06	0.757811623
R17	0.749892873
R11	0.71349671
R14	0.538069432
R10	0.5
R07	0.499881988
R08	0.499256084
R09	0.439815047
R02	0.439684604
R19	0.40986065
R12	0.384295753
R13	0.384295753
R04	0.367234563
R05	0.366306045

R15	0.321484261
R18	0.320972989
R16	0.297244465

Table 7: Ranking of risks

Recommendations

The deterministic probability of a perfect order is around 98%, being 93.34% in the worst 5% of cases. This indicates a solid supply chain for mass commercialization while maintaining quality. Still, this percentage is lower when considering HPIs that require a "white glove" approach, with a more concrete review of the different processes to ensure their satisfaction. Having managers assigned to specific HPI accounts can strengthen relationships and improve trust in the product and brand.

On the other hand, to maintain the high probability of a perfect order, it is necessary to modernize and digitize the supply chain to improve resilience and response time (Belhadi et al., 2021). Additionally, regarding social risk, transparent supply chains are a necessity because a bad partner can cause a catastrophic reputational hit (Diego and Montes-Sancho, 2025). Finally, continuous risk assessment will ensure that appropriate measures are taken to maintain the competitive advantage and mitigate possible disruptions to the business.

Disaster-Recovery Architecture

Pamper Pets requires an application architecture that supports a 1-minute RPO, 1-minute RTO, and 1-minute changeover. To make this possible, it is necessary to use multiple cloud providers, as their maximum availability SLAs are 99.99%, resulting in 52.6 minutes of downtime per year. Additionally, there have been various incidents involving cloud service providers (CSPs) that have highlighted the problems of relying on a single provider (Kleinman, 2025).

The following assumptions were made for cloud architecture:

- The presented architecture shows the production environment, but development and staging are not presented

The proposed architecture uses a global load balancer (GSLB) to distribute traffic across two active-active CSPs in multiple availability zones. GSBL monitors the service across both CSPs and allows clients to connect to the application by changing DNS in case one CSP experiences availability issues. The proposed time between monitor connections is 30 seconds with four retries every 5 seconds. The next step in both CSPs is the load balancer, which distributes traffic to the front-end machines. It is recommended to use the CSP's load balancer to reduce management costs, such as updates. Depending on the level of control needed, it could use a PaaS instead of IaaS for scaling. This design choice allows for high availability on the front end, with up to 99.99% (Amazon Web Services, 2022a; Microsoft, 2025) for the load balancer and 99.99% (Amazon Web Services, 2022b; Microsoft, 2025) for the application hosted in compute services. The next step follows the same principle: use a load balancer for a database farm. In this configuration, the load balancer maintains 99.95% availability (Amazon Web Services, 2022a; Microsoft, 2025), and the database retains up to 99.95% availability (Amazon Web Services, 2024a; Microsoft, 2025).

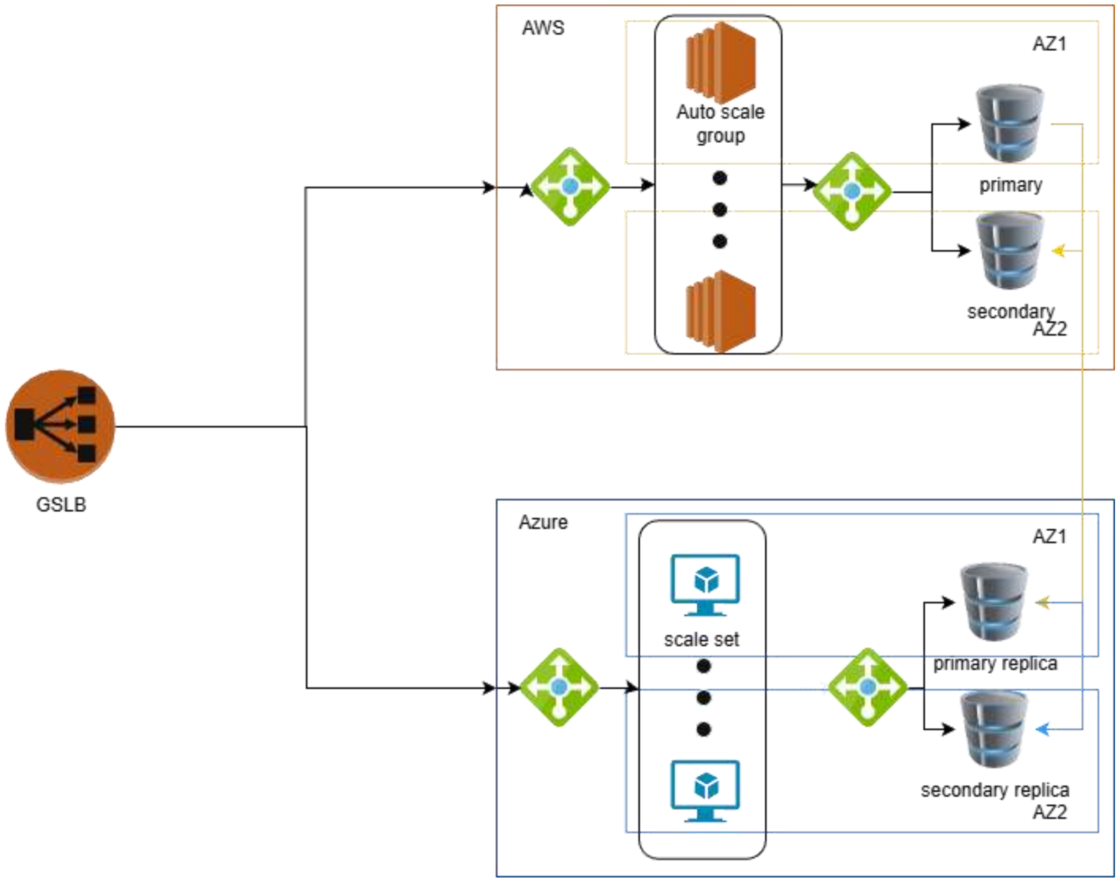

Table 8: Proposed architecture

To obtain the availability (A) in one CSP, we need to calculate it considering the hard dependencies (Amazon Web Services, 2024):

$$A = 99.99\% \times 99.99\% \times 99.99\% \times 99.95\%$$

$$A = 99.88\%$$

Now, in this case, we have a redundant system, we can get the unavailability UA by using the following:

$$UA = (1 - 0.9988)\% \times (1 - 0.9988)\%$$

$$UA = 0.12\% \times 0.12\%$$

$$A = 0.000144\%$$

Over a year, this represents 0.33 minutes of downtime, which is right on the RTO and RPO required for the system. To maintain GDPR compliance, it is essential to ensure that CSP tenants are provisioned within approved geographical regions across the European Union.

Vendor-Lock-in Mitigation

The proposed architecture mitigates vendor-lock-in by following best practices:

- Use open-source technologies for data (PostgreSQL or MariaDB) and application (Kubernetes) compatibility that are standard and will allow rapid change of CSP if necessary
- Follow best practices of open and tested standards (CIS Benchmark) to avoid depending on the services of the CSPs for compliance
- Use multiple CSPs, avoiding dependence on one provider and distributing the application over various providers, developing a simple cloud exit strategy
- Additionally, it is recommended to use infrastructure as code to have a simple and effective way to deploy the infrastructure in other providers if needed
- Ensure proper disaster recovery testing to ensure a correct understanding of the solutions, the changeover time, and requirements

Conclusion

This report presents specific recommendations to improve the supply chain for massive consumers, HPI, and cloud architecture. The recommendations will ensure a resilient and efficient supply chain and architecture while maintaining reputation and quality.

Appendix 1: code

```
import numpy as np
import matplotlib.pyplot as plt

# Set seed for reproducibility
np.random.seed(42)
iterations = 1000000

def run_risk_simulation():

    r_warehouse = np.random.triangular(0.995, 0.99745, 0.9999, iterations)
    r_quality = np.random.triangular(0.97, 0.98, 0.99, iterations)
    r_shipping = np.random.triangular(0.95, 0.97, 0.99, iterations)
    r_cyber = np.random.triangular(0.9999985, 0.99999853, 0.99999856, iterations)

    r_sys = r_warehouse * r_quality * r_shipping * r_cyber

    mean_res = np.mean(r_sys)
    p5 = np.percentile(r_sys, 5)
    p95 = np.percentile(r_sys, 95)

    print(f"--- Monte Carlo Results (n={iterations}) ---")
    print(f"Mean Reliability: {mean_res:.4f}")
    print(f"5th Percentile (Value at Risk): {p5:.4f}")
    print(f"95th Percentile (Best Case): {p95:.4f}")

    plt.figure(figsize=(12, 6))
    plt.hist(r_sys, bins=50, color='skyblue', edgecolor='black', alpha=0.7, density=True)

    plt.axvline(mean_res, color='red', linestyle='dashed', linewidth=2, label=f'Mean: {mean_res:.2%}')
    plt.axvline(p5, color='orange', linestyle='dotted', linewidth=2, label=f'5% VaR: {p5:.2%}')

    plt.title('Monte Carlo Simulation: Cathy’s Digital Transformation Success Probability', fontsize=14)
    plt.xlabel('Probability of "Perfect Order" Success ($R_{sys}$)', fontsize=12)
    plt.ylabel('Frequency', fontsize=12)
    plt.legend()
    plt.grid(axis='y', alpha=0.3)
    plt.show()

if __name__ == "__main__":
    run_risk_simulation()
```

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